# IGCSE 0607 Practice Questions — E1.3

## Powers and roots

**Total Questions:** 15 — **Total Marks:** 80 Marks

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**1** **(a)** Write down the value of $\sqrt{121}$.

..... **[1]**

**(b)** Work out the value of

$$4^3 - \sqrt[3]{27}$$

..... **[2]**

**2** **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

**(b)** Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

**3** **(a)** Work out the value of

$3^3 - \sqrt{144}$

..... **[2]**

**(b)** Work out the value of

$2^4 \times \sqrt[3]{125}$

..... **[2]**

**4** **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

**(b)** Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

**(c)** Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

**5** **(a)** Work out the value of

$5^2 - \sqrt{121}$

..... **[1]**

**(b)** Work out the value of

$\sqrt[3]{27} \times 2^4$

..... **[2]**

**(c)** Work out the exact value of

$$\frac{6^2 + \sqrt{64}}{\sqrt[3]{125}}$$

..... **[2]**

**6** **(a)** Work out the exact value of

$$\frac{5^2 + \sqrt{144}}{2^3 - \sqrt[3]{1}}$$

..... **[2]**

**(b)** Find the value of $k$ when

$$3^k = \sqrt{81} \times 3^2$$

$k =$ ..... **[2]**

**7** **(a)** Calculate the value of

$$\frac{4^3 + \sqrt{225}}{\sqrt[3]{64}}$$

Give your answer as a decimal.

..... **[3]**

**(b)** Given that $p = 5$ and $q = -2$, work out the value of

$$p^2 - q^3$$

..... **[2]**

**8** **(a)** A cube has a volume of 125 $cm^3$.
Find the length of one edge of the cube.

..... cm **[1]**

**(b)** Work out the total surface area of this cube.

..... $cm^2$ **[2]**

**(c)** Given that $\sqrt{N} = 3^3 - 4^2$, find the value of $N$.

$N =$ ..... **[2]**

**9** Solve the following equations.

**(a)**

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

**(b)**

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

**(c)**

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

**10** **(a)** Find the integer $k$ such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

**(b)** Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

**(c)** Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of $y^2$.

..... **[3]**

**11** **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

**(b)** Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

**(c)** Find the largest integer $n$ such that $n^4 < 2000$.

$n =$ ..... **[2]**

**12** **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

**(ii)** Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

**(b)** Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

**13** **(a)** Calculate the value of

$$\frac{\sqrt[3]{45.6} + 2.1^4}{3.8^2}$$ Give your answer correct to 3 significant figures.

..... **[2]**

**(b)** A solid cube has a volume of 4913 $cm^3$.
Calculate the total surface area of the cube.

..... $cm^2$ **[3]**

**14** **(a)** Calculate the value of

$\sqrt[5]{7.59375}$

..... **[1]**

**(b)** Find the two possible values of $x$ in the equation

$2x^4 - 17 = 1233$

$x =$ ..... , $x =$ ..... **[3]**

**(c)** It is given that

$$y = \sqrt{\frac{3a^2 + b}{c}}$$

Work out the value of $y$ when $a = 4.2$, $b = -5.7$ and $c = 1.8$.
Give your answer correct to 2 decimal places.

$y =$ ..... **[3]**

**15** **(a)** Calculate the value of

$$\frac{\sqrt[4]{150000}}{4.5^3 - \sqrt{75}}$$ Give your answer correct to 3 decimal places.

..... **[2]**

**(b)** $x$ is a positive integer such that $\sqrt[3]{x^2} = 16$.
Find the value of $x$.

$x =$ ..... **[3]**

**(c)** The volume of a sphere is equal to the volume of a cube with edge length 12 cm.
Calculate the radius of the sphere.
The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.

..... cm **[3]**